Keyspark

Real-time detection of input automation over a Kafka + Spark streaming pipeline

Murat Emirhan Aykut

Bahçeşehir University

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PROBLEM AND MOTIVATION

One unbounded input stream, one question

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The data

Every key press, mouse move, click and scroll. An unbounded, high-rate, bursty event stream - data that arrives, not data that sits.

The application

Jigglers, auto-clickers and keep-awake tools fake presence. Flag, per minute, whether input came from a human or a tool.

Why it is a Big Data problem

Event-time windowing, out-of-order arrival, producer vs consumer rate mismatch, fault tolerance, and a fast-vs-complete split.

Privacy by construction

Only key identifiers and timings are kept. No raw typed text is ever persisted - the whole signal is content-free.

The pipeline is the contribution. The detector is the application that gives the stream a purpose.

RELATED WORK

Streaming systems meet behavioral signals

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Streaming foundations

Kafka: the replayable commit log *Kreps 2011*

Spark RDDs: fault-tolerant engine *Zaharia 2012*

Structured Streaming: stream as a growing table *Armbrust 2018*

Lambda architecture: speed + batch *Marz 2015*

Streams of interaction data

Kafka pipeline over a browser clickstream *Pal 2023*

Same Kafka + Spark Lambda stack, other domain *Demirezen 2023*

TimeAware: per-time-bin self-monitoring *Kim 2016*

Keystroke + mouse dynamics

Keystroke timing as a behavioral biometric *Monrose 2000*

Emotional state from keystroke timing *Epp 2011*

Mental fatigue from keystroke dynamics *Acien 2022*

Detecting automation

Passive game-bot detection from input *Gianvecchio 2009*

Human vs bot vs cyborg accounts *Chu 2012*

BeCAPTCHA-Mouse: synthetic bot trajectories *Acien 2022*

KeySpark's niche: a live streaming pipeline computing a content-free human-vs-automation signal over combined keyboard + mouse events.

DATASET AND VISUALIZATION

A self-collected, content-free event archive

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~1.0 x 10^6

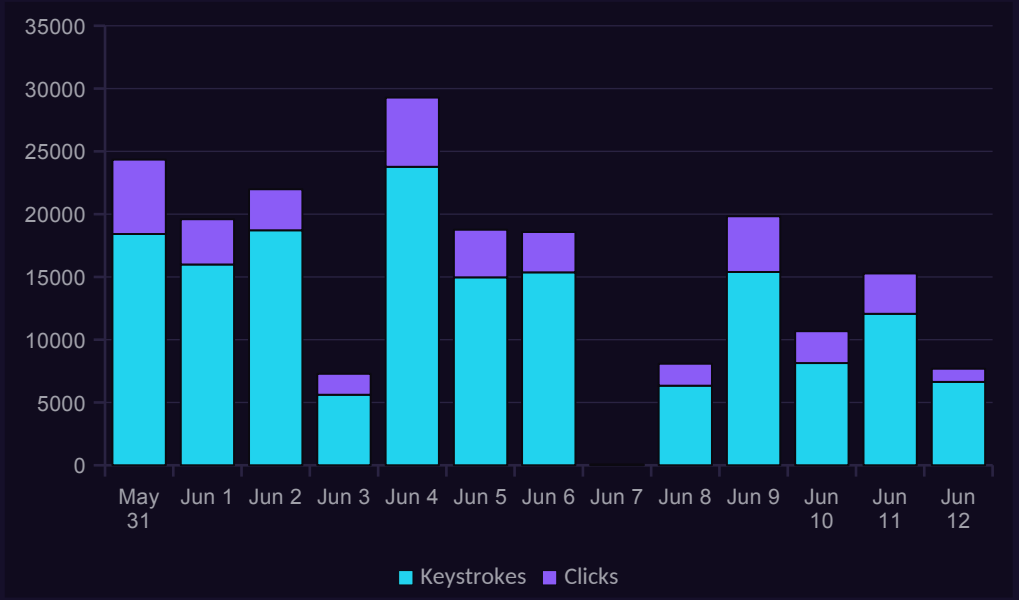
events, ~12 days, one user, one machine

preliminary - data freeze pending

- Key identifiers + timings only. No raw text is ever stored.
- Mouse moves throttled to ~50/s. Clicks and scrolls never throttled.
- JSON events parsed with an explicit schema, stored as Parquet.

```
type key x y button pressed dx dy user ts
```

Daily captured activity (from output/metrics, May 31 - Jun 12)



Real archive data. Words and corrections are counted from the same keystrokes; ~50/s throttled mouse moves are archived but not shown.

SYSTEM ARCHITECTURE

One stream in, three kinds of truth out

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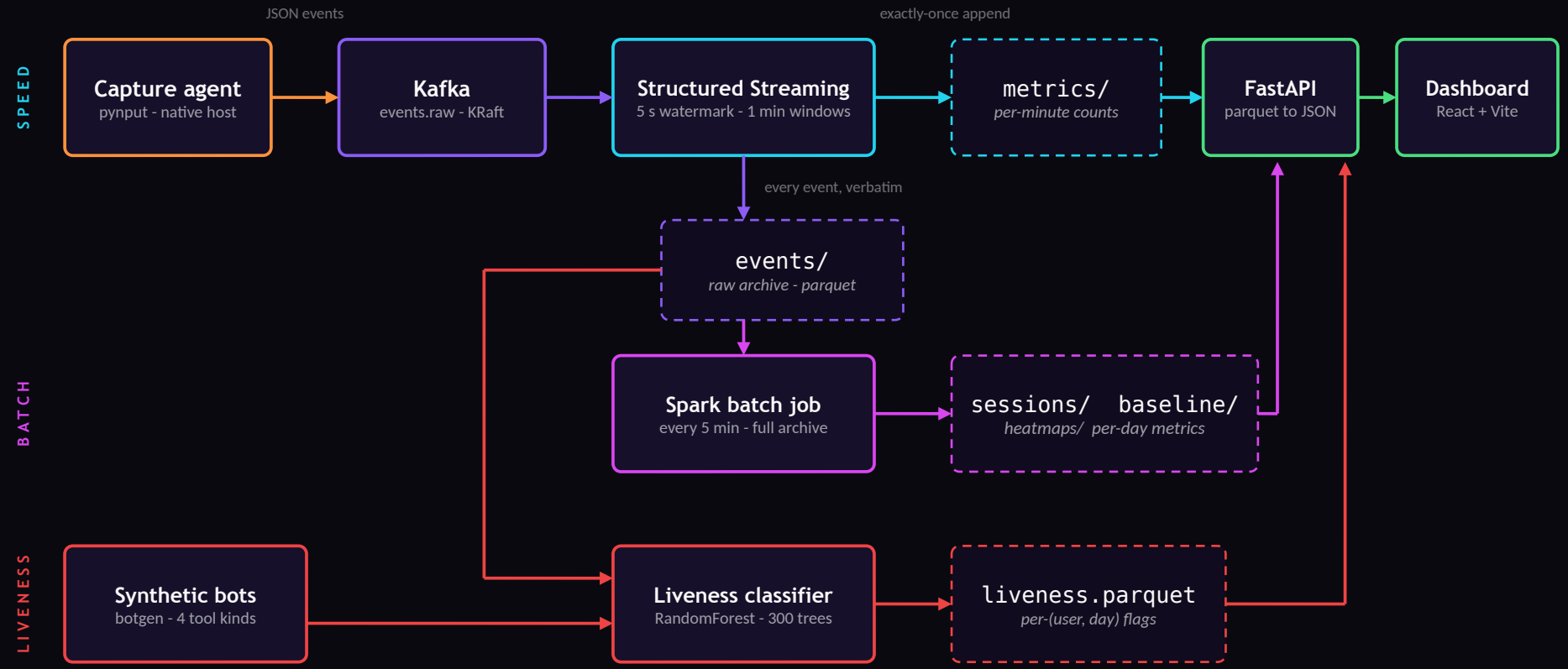

solid = process dashed = Parquet store **speed** / **batch** / **liveness**

TECHNOLOGY JUSTIFICATION

Why this stack - every piece earns its place

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Apache Kafka

ingestion log

A durable, replayable, ordered log decouples the agent from Spark: capture never blocks, bursts are buffered, and offsets are what make replay and exactly-once possible. A direct DB write or a fire-and-forget queue gives none of this.

Structured Streaming

speed layer

Event-time windows plus a watermark stay correct under late, out-of-order arrival - a naive consumer loop cannot. Checkpointed offsets and atomic Parquet commits give exactly-once. The same code runs on local[*] and on a cluster.

Spark batch

batch layer - Lambda

Sessions and inter-event timing depend on the previous event; lag() cannot see across micro-batches. So the stream lands everything, and a batch pass over the bounded archive is the source of truth. That split is the Lambda architecture, by name.

Parquet

serving store

Columnar, compressed, splittable - shaped for the dashboard's analytical scans. It is the file sink Spark commits to exactly-once, and it adds zero services to operate.

THE EXAM QUESTION

Why not just write events to MongoDB?

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What MongoDB is built for

- Point reads and writes of mutable documents
- Secondary indexes for selective lookups
- Flexible per-document schema
- Genuinely simpler to stand up for low-rate CRUD

What this workload is

- Append-only, high-rate event stream - nothing is ever updated
- Windowed aggregation over everything, not point lookups - columnar scans win
- Needs event-time windows, watermarks, exactly-once - Mongo has none; we would rebuild them in app code
- Needs replay for recovery and reprocessing - the Kafka offset log gives it for free

Not a bad database - the wrong shape. On Mongo we would hand-build windowing, late-event handling, exactly-once and replay in application code. Kafka + Spark + Parquet give exactly those, natively. The honest flip side: for low-rate mutable CRUD, Mongo would be the simpler and better choice.

Event-time windows and the 5 s watermark

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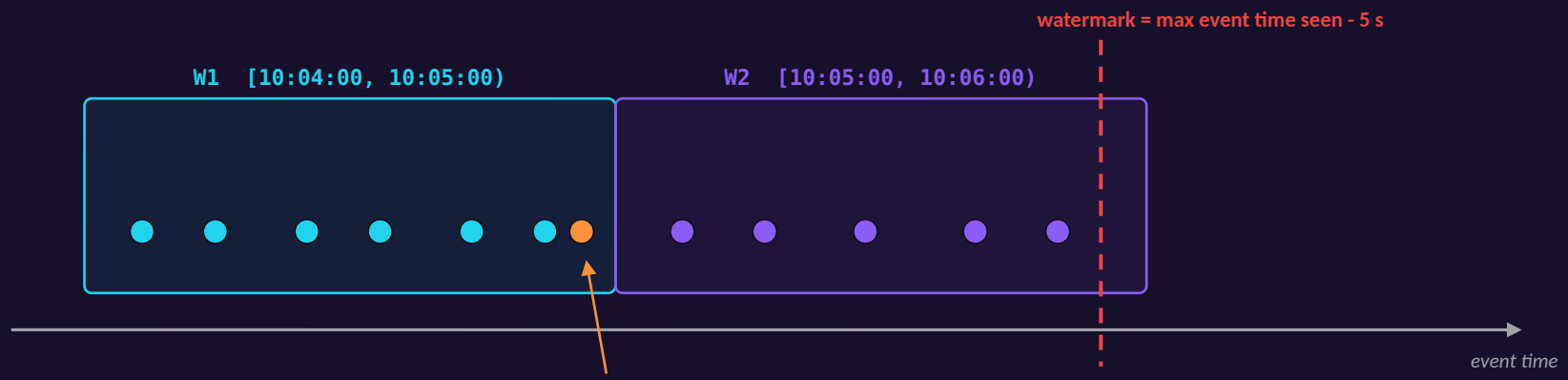
`event time` when the key was pressed (ts in the event) - windows are defined on this

`processing time` when Spark happens to see it - irrelevant to the result



Why the watermark is mandatory: the Parquet sink is append-only - each window's row is emitted exactly once, never rewritten. Without a watermark Spark could never declare a window closed, so append could emit nothing.

One stream, two queries, exactly-once on disk

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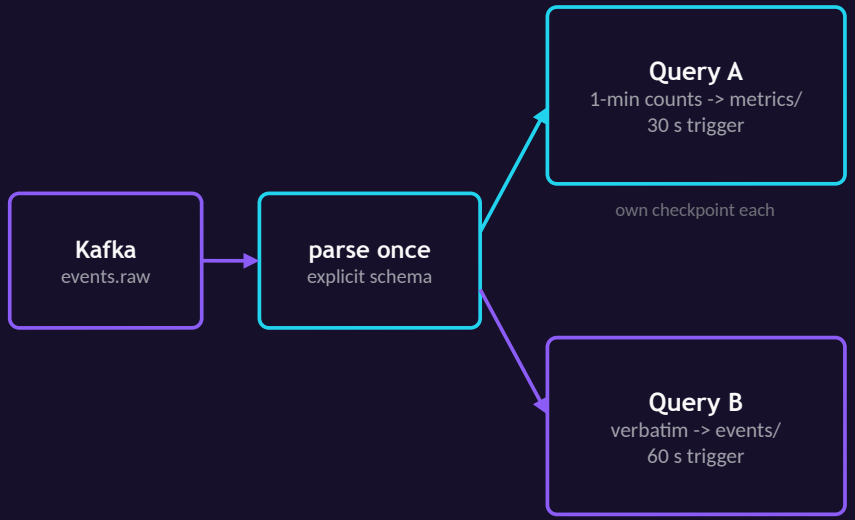
One parsed stream, two sinks



Without triggers, every micro-batch wrote a part file (100k+ tiny files once).

Exactly-once, end to end

`Kafka offsets` consumed range recorded per micro-batch

↓

`Checkpoint (WAL + state)` offsets and window state survive restarts

↓

`Atomic Parquet commit` _spark_metadata manifest - a file exists only once committed

↓

`Restart = resume` reprocess from offsets; never lose, never duplicate

Sleep/wake hardening: failOnDataLoss=false resets to the earliest offset instead of crash-looping; restart loops + a watchdog bounce only the wedged component.

THE APPLICATION

Liveness: was this minute typed by a human?

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7 cross-modal shape features (per 1-min window)

- **key_diversity** distinct keys / keystrokes
- **ks_iei_cv** inter-keystroke interval CV
- **move_iei_cv** inter-mouse-move interval CV
- **iei_cv** all-event cadence CV
- **step_mean / step_cv** mouse step - rigid for a jiggler
- **mouse_fraction** near 1 jiggler, near 0 key tool

Shape, not volume: regularity and variety, never content. Windows with < 5 events abstain.

RandomForest - 300 trees, balanced classes

Human class = the real archive. Non-human class = synthetic events from botgen: a fixed-pattern jiggler, an F15-style keep-awake, an auto-typer, an auto-clicker - cadence jittered so only the robotic shape remains. One featurizer for both classes: no train/serve skew.

A day turns red when >= 2 windows score >= 0.8

a sustained burst of automation, not one odd minute - surfaced on the dashboard calendar after each 5-min batch refresh

Honest caveat: the non-human class is synthetic. Validation on real captured automation is named future work.

RESULTS AND DISCUSSION

Throughput first, the classifier second

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PRELIMINARY - DATA FREEZE PENDING

Pipeline performance - single machine, local[*]

Layer	Throughput	Latency
Spark batch (full pipeline)	~7.0 x 10^5 ev/s	-
Structured Streaming	~2.0 x 10^5 rows/s	275 ms / micro-batch
End-to-end (keypress to UI)	-	~65 s

~65 s = 60 s window + 5 s watermark. A per-minute metric cannot exist before its minute ends - the floor is the design, not the engine. Measured from Spark's own StreamingQueryProgress.

Liveness - stratified hold-out

Accuracy	0.991
Precision	0.976
Recall	0.994
F1	0.985
ROC-AUC	0.9997

Top features: iei_cv, move_iei_cv, step_cv - the regularity tells.

Read the right-hand numbers carefully: the non-human class is synthetic, so the hold-out partly measures separation from our own generator. We claim the features are discriminative - we do not claim a field detection rate. Validation on real captured automation is future work.

DEMONSTRATION

The live demo, right after these slides

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1 - Type

I type; within ~65 s the per-minute metrics, time series and heatmap update on the dashboard.

watch the window + watermark latency happen

2 - Inject a jiggler

botgen pushes synthetic jiggler events into Kafka; a sustained burst scores non-human and the calendar day flips red.

the whole pipeline reacts end to end

3 - Open the Spark UI

the Structured Streaming tab shows live input rate vs processing rate and micro-batch durations.

the engine's own evidence, not mine

Then a walk through the code: the streaming job, the batch job, and the featurizer.

LIMITATIONS AND FUTURE WORK

What this is not, yet

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Honest limits

- Single broker, single user, single machine (local[*]) - partitioning and shuffle never really stressed
- Synthetic non-human class - hold-out scores are inflated relative to automation in the wild
- Idle suppressors (caffeinate, Amphetamine, PowerToys Awake) emit no input events - invisible by construction
- ~65 s latency floor is tied to the 1-min window; shorter windows = noisier labels
- A sleep that kills Spark's local RPC costs a cold restart, not seconds

Future work

- Validate on real captured automation: a hardware USB jiggler emits genuine HID events the agent can capture (software injection is dropped by macOS)
- Ground botgen against the full spectrum of real keep-active tools
- Out-of-stream probe of OS power assertions to catch idle suppressors
- Multi-user ingestion to exercise Kafka partitioning; a real cluster to exercise Spark shuffle

Conclusion

Standard big-data parts, end to end: pynput -> Kafka -> Spark -> Parquet -> FastAPI -> React.

The Lambda split is earned, not decorative: cross-batch ordering forces a batch source of truth.

Streaming correctness is explicit: event time, a 5 s watermark, append + checkpoints = exactly-once.

The ML rides the pipeline and is reported honestly: content-free features, synthetic-class caveat named.

Thank you.

Questions welcome - live demo next.

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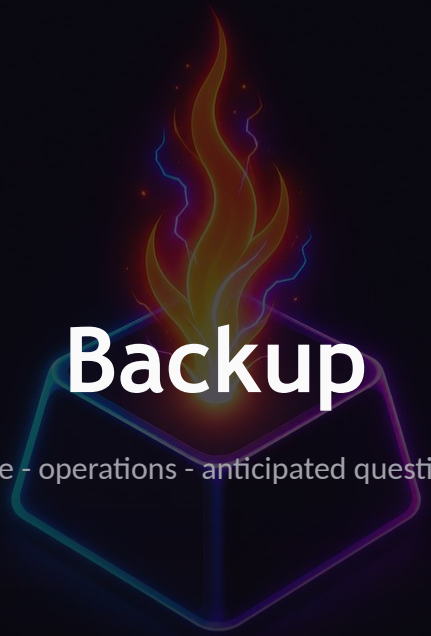

Backup

code - operations - anticipated questions

BACKUP - THE ACTUAL CODE

keyspark/streaming_job.py - parse, window, sink

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```
parsed = (
    raw.selectExpr("CAST(value AS STRING) AS json")
    .select(from_json(col("json"), event_schema()).alias("e"))
    .select("e.*")
    .withColumn("event_time", col("ts").cast("timestamp"))
    .withWatermark("event_time", WATERMARK)   # 5 seconds
)

metrics = (
    parsed.groupBy(window(col("event_time"),
                          WINDOW_DURATION),     # 1 minute
                   col("user"))
    .agg(*event_count_exprs())  # keystrokes, words,
)                               # corrections, clicks

metrics_query = (
    metrics.writeStream.format("parquet")
    .option("path", METRICS_PATH)
    .option("checkpointLocation", METRICS_CHECKPOINT)
    .outputMode("append")
    .trigger(processingTime="30 seconds")
    .start()
)
```

explicit schema
no inference - predictable, cheap, typed

event-time watermark
bounds lateness; makes append legal on a windowed aggregate

tumbling window
per-(user, minute) groups on event time

checkpoint + append + trigger
exactly-once output; trigger caps file creation (30 s metrics, 60 s archive)

BACKUP - OPERATIONAL ANATOMY

Checkpoints, sleep/wake, batch, serving

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Checkpoint + commit anatomy

Per query: Kafka offset ranges (WAL) + window state. The Parquet sink's _spark_metadata manifest commits files atomically - readers only see committed files, so restart resumes without loss or duplicates.

Sleep/wake hardening

failOnDataLoss=false resets truncated offsets to earliest instead of crash-looping. Every process runs in a restart loop; a watchdog listens for the OS wake notification, polls output freshness, probes the Spark RPC, and bounces only the wedged pane.

Batch internals

Every 5 min from an in-process scheduler in the API. Reads the full archive by file glob - independent of the streaming commit log. Sessions via gap-and-island: order by event time per user, flag gaps > 5 min, cumulative-sum the flags into session ids.

Serving

FastAPI reads Parquet and serves JSON; React polls. The calendar joins per-day metrics with liveness flags; a flagged day renders red and drills down to that day's timeline and heatmap.

BACKUP - ANTICIPATED QUESTIONS

Likely probes, prepared answers

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Why not Flink or Kafka Streams?

Lambda needs a batch engine anyway - Spark is one engine and one API for both layers. Flink's lower per-event latency would not help: the floor here is the 1-minute window itself.

Why a 5 s watermark?

One local broker produces little disorder, and the watermark delays emission: at 30 s the first result took ~90 s. 5 s covers observed lateness and keeps the dashboard live.

Why 1-minute tumbling windows?

The product unit is a minute of activity. Shorter = lower latency but noisier liveness labels; longer = smoother but staler.

Why a RandomForest, not a deep model?

7 tabular features and modest data. It retrains in seconds each batch cycle, resists overfitting with balanced weights, and its feature importances (the CV features dominate) explain its decisions.

How exactly is exactly-once achieved?

Offsets + window state checkpointed per micro-batch; the file sink commits atomically via _spark_metadata. Replay after failure reprocesses the same offsets; the sink never double-commits.

Could the agent write to Spark directly?

Then capture blocks on every consumer hiccup and nothing is replayable. The log decouples rates, buffers bursts, and is what makes restart and reprocessing free.